Adarsh Jha

adarshjhaxif@gmail.com | Github | LinkedIn | Portfolio

Education

Heritage Institute of Technology Kolkata, West Bengal
Bachelor of Technology in Computer Science and Engineering — CGPA: 9.34 *2022 – 2026*

Experience

Software Engineer Sept 2024 – Current
Humane Intelligence *Work from Home*

- Developed and deployed various Red-Teaming projects on the platform
- Implemented multi-model orchestration across LLM providers
- Built AI red-teaming features based on client requirements for clients including **NIST, IMDA, UNESCO, DoD**
- Tech stack: Next.js, Vercel AI SDK, PostgreSQL, Docker, AWS

Software Development Engineer Intern Feb 2024 – May 2024
1 Percent Verse *Work from Home*

- Built a React + TypeScript frontend for an AI-powered career roadmap platform (similar to roadmap.sh)
- Translated Figma designs into production UI, rendering AI-generated learning paths and skill trees
- Designed Supabase-backed REST APIs for user data, progress tracking, and roadmap delivery
- Owned the integration layer wiring AI-generated content into the user-facing application

Projects

Patently | *Next.js, TypeScript, FastAPI, Qdrant, Gemini, Cloud Run* | Live | Repo Sept 2026

- Built a prior-art analyzer mapping which of 258,933 patents teach each claim element
- Designed multi-angle retrieval fusing 4–6 query embeddings via Reciprocal Rank Fusion
- Implemented verbatim quote grounding that demotes unverifiable citations; deterministic §103 pair detection
- Fit 258k vectors into a 1 GB free tier via int8 quantization (1.06 GB → 265 MB)

DiffWatch | *Next.js, TypeScript, Gemini AI, Supabase* | Live | Repo April 2026

- Built an AI-powered PR review platform generating actionable feedback in under 15s
- Designed a parallel analysis pipeline for diffs, CI status, vulnerabilities, and bundle impact
- Implemented a multi-persona review system (Architect, Security, PM, Intern) with streaming output
- Added code insights: commit history analysis, risk detection, and anti-pattern flagging

MetaGuard AI | *Next.js, TypeScript, OpenMetadata, Gemini AI* | Live | Repo April 2026

- Built an AI-powered data governance copilot for any Postgres, MySQL, or OpenMetadata catalog
- Engineered a three-signal PII classifier (name heuristics, value patterns, Gemini) with tag write-back to OpenMetadata
- Designed a 0–100 governance score from documentation, tagging, key, and relationship coverage, with ranked gaps
- Built natural-language schema Q&A over read-only connections; credentials stay client-side, no row data reaches the model

Technical Skills

AI / LLM: LangChain, LangGraph, LangSmith, MCP, RAG, Prompt Engineering
Languages: TypeScript, JavaScript, Python, Go, C++
Frontend: Next.js, React, Streamlit, Tailwind, Shadcn, Framer Motion
Backend: Node, Express, FastAPI, tRPC, WebSockets, GraphQL
Databases: PostgreSQL, MongoDB, MySQL, Redis
Infra: AWS, Docker, Kubernetes, Vercel, GitHub Actions, Nginx